

SC-SCOPE endpoint floor sharpening – the reconciliation (proved) and

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

The SC-SCOPE all-orders endpoint ($I = 2 \times 10^{-3}$) did not close by third-order means: the joint incompatible-pairing recovered only $\times 0.905$ (sscope-joint-pairing v1.0), which named the SOLE remaining route as “a larger second-order STEP-5B endpoint floor: if the de-thinned floor were $\gtrsim \times 3.9$ the paired joint would close.” This note completes that route by observing that the floor $\rho = 2.58$ rests on a LOOSE additive-energy bound, and the additive-energy result of the DR-2 programme (R-025/R-026) supplies the tighter constant.

2. The floor composition

The de-thinned STEP-5B closure floor (robustness_mu2_step5b_remargin) is

$$\rho = \frac{K_{\text{budget}}}{K(n_{\text{pack}})}, \quad K_{\text{budget}} = \frac{4(1 - a_0)m}{(\lambda'I)^2 J_{\text{eff}}}, \quad K(n_{\text{pack}}) = 8 + 4\sqrt{14}\sqrt{n_{\text{pack}}},$$

where $K(n_{\text{pack}})$ is the kappa-balanced additive-energy bound and $n_{\text{pack}} = 16/\theta_{\min}^2$. At the endpoint (anchor $\mu^2 = 0.005$): $n_{\text{pack}} = 40.7$, $K(n_{\text{pack}}) = 103.5$, $K_{\text{budget}} = 266.7$, so $\rho = 2.58$ (reproducing the AddE $\times 2.6$).

3. The bound is loose at the small endpoint

$K(n) = 8 + c_R\sqrt{n}$ with $c_R = 4\sqrt{14} \approx 15$ is the UNCONDITIONAL triple-count bound ($E_+ \leq n^{5/2} \Rightarrow K = E_+/n^2 \leq \sqrt{n}$ up to the prefactor). At the endpoint $n_{\text{pack}} = 40.7$ this gives 103.5, which EXCEEDS the trivial Lemma-A additive-energy bound

$$\frac{E_+(Q)}{N^2} \leq 1 + T'(Q), \quad T'(Q) \leq |Q| \leq n_{\text{pack}} = 40.7 \quad \Rightarrow \quad 1 + T' \leq 41.7 < 103.5.$$

The large prefactor $c_R \approx 15$ makes the $\sqrt{n}$ bound overshoot the trivial $O(n)$ bound at this moderate n . The additive-energy constant K and $1 + T'$ are the SAME λ' -free quantity $(\langle F^4 \rangle - 4I^2)/I^2$ (via $\sum w^2 = \lambda'^2(\langle F^4 \rangle - 4I^2)$ and $\sum w^2 \leq K(\lambda'I)^2$), so $1 + T'$ is a legitimate, tighter substitute.

4. The sharpened floor: a partial advance, NOT a closure

Correction (v1.3). v1.1/v1.2 here computed the closure as $\text{paired} = \rho_{\text{lat}}/(1 + \max[R_s + R_q]) = \rho_{\text{lat}}/2.872$ and concluded the endpoint LIFTS. That formula’s linear-in- ρ growth is only a LOCAL approximation at $\rho = 2.6$. The physically-correct accounting (`scscope_joint_endpoint`) treats the third-cumulant SUNSET as an ABSOLUTE cost $C_{\text{sunset}} = \text{composed}/1.13$ that does NOT vanish as the second-order floor thickens; the joint ratio therefore SATURATES:

$$\text{joint}(\rho) = \frac{\text{MARGIN}}{C_2 + C_{\text{sunset}} + C_{\text{quartic}}} \xrightarrow{\rho \rightarrow \infty} \times 1.13,$$

NOT linearly. At the sharpened floor ($K \leq T' \leq n_{\text{pack}} = 40.7 \Rightarrow \rho_{\text{lat}} = 6.55$) the additive joint is only $\times 0.945 < 1$; the verified $K \leq 0.52 T'$ gives $\rho_{\text{lat}} = 12.6$ and $\times 1.026$ – MARGINAL. The threshold for $\text{joint} > 1$ is $\rho \geq 9.85$ (i.e. $K \leq 27$), not the 3.9 the lift used (`scscope_joint_correction.py`, 5/5). So the proved floor sharpening is a real PARTIAL advance – it moves the additive endpoint joint from $\times 0.757$ to $\times 0.95$ – 1.03 – but does NOT cleanly close the endpoint.

5. Reconciliation (PROVED); the lift retracted

The constant map is now pinned EXACTLY. In the STEP-5B convention $w_t = \lambda' \sum_{u+v=t} A_u A_v$ and, by the Parseval reformulation, $\sum_{t \neq 0} w_t^2 = \lambda'^2 (\langle F^4 \rangle - 4I^2)$, so the floor’s additive-energy constant is exactly $K = \sum_{t \neq 0} w_t^2 / (\lambda' I)^2$. Writing $v_t = \sum_{u+v=t} A_u A_v$ (so $w_t = \lambda' v_t$), the weighted Lemma A (R-027) applied to the λ' -free v_t gives, for its $t \neq 0$ part, $\sum_{t \neq 0} |v_t|^2 \leq T'(M) I^2$; hence $\sum_{t \neq 0} |w_t|^2 = \lambda'^2 \sum_{t \neq 0} |v_t|^2 \leq T'(M) (\lambda' I)^2$ and

$$K = \frac{\sum_{t \neq 0} w_t^2}{(\lambda' I)^2} \leq T'(M)$$

where $T'(M)$ is the sum-circle richness of the mode set – NOT the looser kappa-balanced $8 + c_R \sqrt{n}$. Verified (`scscope_constant_map.py`, 3/3) on real $\pm k$ -symmetric sphere-lattice patterns: $K_{\text{floor}}/T' \leq 0.52$ and $w_0 = I$ exactly (so the $-4I^2$ subtraction is even more conservative, lowering K further). With $K \leq T' \leq n_{\text{pack}} = 40.7$ (separated) or $T' \sim \text{tens}$ (lattice), $\rho_{\text{lat}} = K_{\text{budget}}/K \geq 266.7/40.7 = 6.55$. This is a real, PROVED sharpening of the second-order floor; but per §4 it does NOT close the endpoint under the correct additive bookkeeping (it is marginal, $\times 0.95$ – 1.03). The reconciliation $K \leq T'(M)$ STANDS; the LIFT does not.

Grade (v1.3 corrected). The reconciliation $K \leq T'(M)$ is PROVED and the second-order floor is genuinely sharpened ($\rho : 2.58 \rightarrow 6.55$). But §4 shows this does NOT close the all-orders endpoint: under the correct additive bookkeeping the sunset saturates the joint at $\times 1.13$, and at the sharpened floor the joint is only $\times 0.945$ – 1.026 (marginal). The LIFT claimed in v1.1/v1.2 is therefore RETRACTED. SC-SCOPE is RESTORED as a B1 named hypothesis ($\{\text{H-LAYER}\} \rightarrow \{\text{H-LAYER}, \text{SC-SCOPE}\}$), B1 tier UNCHANGED T6. This note’s durable content is the PROVED floor sharpening $K \leq T'(M)$ and the resulting PARTIAL advance ($\times 0.757 \rightarrow \times 0.95$ – 1.03); a clean lift needs either $\rho \geq 9.85$ ($K \leq 27$, tighter than $n_{\text{pack}} = 40.7$) or a rigorous third-cumulant bound beating the saturating sunset.

5b. The realized quartic (strong evidence, not certified)

Under the canonical additive bookkeeping the endpoint closes at $\rho_{\text{lat}} = 6.55$ iff the quartic-difference per-transfer ratio satisfies $R_{\text{max}} < 0.634$ (the sunset is rigorous and caps the joint at $\times 1.13$). The

prior $R_{\max} = 1.019$ inherited the Young-ceiling estimate $R_{\sup} = 1.59$ (flagged “not recomputed”). Evaluating the realized ratio DIRECTLY,

$$R(t) = 12 \left(\frac{5v}{2} \right)^2 \lambda'^{-2} \widehat{G^4}(t) \frac{4(1-a_0)}{J(t)}, \quad \widehat{G^4} = (J * J)(t),$$

on the endpoint chords (scscope_quartic_realized.py, 4/4) gives

$$R_{\max}^{\text{realized}} \approx 0.385 < 0.634$$

– the Young ceiling was loose by $\sim 2.6\times$. This is **STRONG EVIDENCE** that the SC-SCOPE all-orders endpoint CLOSES. It is NOT certified: the absolute $\widehat{G^4}$ normalisation carries a factor- $2/(2\pi)^3$ convention (the same class as the $M' = -J(0)$ vs $-J(0)/2$ slip), which is load-bearing here – the closure survives a +50% slack ($0.385 \times 1.5 = 0.577 < 0.634$) but NOT a full factor 2 ($0.769 > 0.634$). The convolution SHAPE is rigorous (grid-converged); only the absolute normalisation is open. NO lift is enacted; SC-SCOPE stays a B1 named hypothesis pending the convention being pinned.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “*The reconciliation $K \leq T'(M)$ might be wrong.*” DISMISSED: it is proved (Parseval + Lemma A $t \neq 0$ part) and verified ($K_{\text{floor}}/T' \leq 0.52$, $w_0 = I$; scscope_constant_map.py 3/3). This is the note’s durable content and is NOT retracted.
- (β) “*Then why is the endpoint not closed?*” Because the third-order joint is SUNSET-LIMITED. Under the canonical additive bookkeeping the sunset cost $C_{\text{sunset}} = \text{composed}/1.13$ does not shrink with the floor, so $\text{joint}(\rho) \rightarrow \times 1.13$ as $\rho \rightarrow \infty$. At $\rho_{\text{lat}} = 6.55$ the joint is only $\times 0.945$ (with the current quartic $R_{\max} = 1.019$); even a perfect quartic gives at most $\times 1.108$. So the floor sharpening alone cannot close it – it is a partial advance.
- (γ) “*The earlier paired= $\rho/2.872$ closure was wrong.*” CONCEDED – this is the v1.3 retraction. That formula’s linear-in- ρ growth is a local approximation at $\rho = 2.6$; the physically-correct accounting (the third-order sunset is an absolute cost) saturates. The retraction is logged at negative-results AUDIT-2026-06-08-scscope-lift-overclaim.
- (δ) “*Is $\rho = 2.58$ reproduced?*” Verified: $K_{\text{budget}}/K(n_{\text{pack}}) = 266.7/103.5 = 2.58$ matches the AddE/remargin $\times 2.6$.

Quantitative sanity check (mandatory). Reconciliation: $K \leq T'(M)$, $K_{\text{floor}}/T' \leq 0.52$, $w_0 = I$ (scscope_constant_map.py 3/3). Floor: $\rho : 2.58 \rightarrow 6.55$ (proved sharpening). Joint (CORRECTED, additive bookkeeping): $\times 0.945$ at $\rho_{\text{lat}} = 6.55$; threshold $\rho \geq 9.85$; saturates at $\times 1.13$ (sunset-limited); the quartic must reach $R_{\max} < 0.634$ to close at $\rho_{\text{lat}} = 6.55$ (scscope_joint_correction.py 5/5). Direction: tighter $K \Rightarrow$ larger $\rho \Rightarrow$ smaller C_2 only; the sunset is the binding term. No closure claimed.

7. Result footer

Result ID: R-029 / SC-SCOPE endpoint floor sharpening (PARTIAL;
lift retracted)

Precise statement: PROVED: the floor's additive-energy constant $K = \sum_{t \neq 0} w_t^{-2} / (\lambda I)^{-2} \leq T'(M)$ (reconciliation, scscope_constant_map.py 3/3), sharpening the endpoint floor rho: 2.58 $\rightarrow$ 6.55 (the kappa-balanced $K=103.5$ was loose at $n_{\text{pack}}=40.7$). RETRACTED: the resulting all-orders LIFT -- it used $\text{paired}=\text{rho}/2.872$ (local approx); the correct additive bookkeeping (sunset absolute) gives only $x0.945$ (conservative) .. $x1.026$ (verified $0.52T'$), MARGINAL; threshold $\text{rho} \geq 9.85$, joint saturates at $x1.13$ (sunset-limited). PARTIAL advance $x0.757 \rightarrow x0.95-1.03$.

Scope: SC-SCOPE all-orders endpoint ($I=2e-3$); the named 'sharper floor $\text{rho} \geq 3.9$ ' route, now completed.

Dependencies: robustness_mu2_step5b_remarg (floor); R-025 Lemma A (additive-energy bound $1+T'$); R-026 (lattice $T' \sim R^{\text{eps}}$); scscope-joint-pairing v1.0 (third-order inflation 2.872).

Evidence grade: Reconciliation $K \leq T'(M)$ PROVED + verified (3/3); floor sharpened rho 2.58 $\rightarrow$ 6.55. The all-orders LIFT is RETRACTED (the additive joint is MARGINAL $x0.945-1.026$, sunset-limited; scscope_joint_correction.py 5/5). Net: a PROVED PARTIAL advance, NOT a closure.

Reproduction command: python codes/vacuum/scscope_floor_sharpening.py

Expected output: scscope_constant_map.py 3/3 ($K \leq T'(M)$); scscope_joint_correction.py 5/5 (additive joint $x0.945$ at $\text{rho}=6.55$; threshold $\text{rho} \geq 9.85$; saturates $x1.13$).

Falsification gate: The reconciliation $K \leq T'(M)$ failing (not observed). The lift is already retracted; closing the endpoint needs $\text{rho} \geq 9.85$ ($K \leq 27$) OR a certified quartic $R_{\text{max}} < 0.634$.

Tier before / after: LIFT RETRACTED. SC-SCOPE RESTORED as a B1 named hypothesis: B1 {H-LAYER} $\rightarrow$ {H-LAYER, SC-SCOPE}, tier UNCHANGED T6. SC-SCOPE endpoint OPEN/MARGINAL. B5 T5. AUDIT-2026-06-08-scscope-lift-overclaim.

No-overclaim statement: Only the reconciliation $K \leq T'(M)$ and the floor sharpening ($\text{rho} 2.58 \rightarrow 6.55$) are claimed (proved). The all-orders endpoint is NOT closed: the additive joint is marginal ($x0.945-1.026$) and sunset-limited (saturates $x1.13$). SC-SCOPE remains a B1 named hypothesis.

Next required action: Pin the absolute Ghat4 normalisation (factor- $2/(2\pi)^3$ convention): the REALIZED quartic $R_{\text{max}} \sim 0.385 < 0.634$ (scscope_quartic_realized.py 4/4) is strong evidence of closure, load-bearing only on that convention. Pinning it (survives +50%, not $x2$) would certify the endpoint lift.